FUNCTIONAL TEST SUITE SPECIFICATION

TicketWave — Functional Test Cases

A detailed breakdown of automated end-to-end booking & payment test scenarios, idempotency guarantees, race condition guards, and coverage gates.

SECTION 1

Test Coverage & Execution Summary

670 passing tests with 0 failures, backed by real PostgreSQL Testcontainers integration testing.

Functional Suite Coverage Summary

Flow Domain	Unit Test Class (Cases)	Integration Test Class (PostgreSQL)	Controller MockMvc Class
Booking Creation & Lifecycle	`BookingServiceImplTest` (44)	`BookingFlowIT` (4)	`BookingControllerTest` (19)
PNR Generation	`PnrGeneratorImplTest` (3)	—	—
Seat Hold & Expiration	`SeatHoldServiceImplTest` (21)	`SeatHoldConcurrencyIT` (1)	—
Payment & Idempotency	`PaymentServiceImplTest` (21)	`PaymentFlowIT` (5)	via `BookingControllerTest`
Reschedule & Settlement	`RescheduleServiceImplTest` (17)	—	via `BookingControllerTest`
Refunds & Policy	`RefundServiceImplTest` (26)	`RefundFlowIT` (8)	`RefundControllerTest` (4)

Booking Creation & Idempotency Rules

Idempotency Key Deduplication

Replayed Key: Submitting an identical `idempotencyKey` returns the original booking without re-holding seats (`BK-02`).

DB Race Resolution: Concurrent inserts with the same key throw `DuplicateBookingRequestException` (`BK-03`).

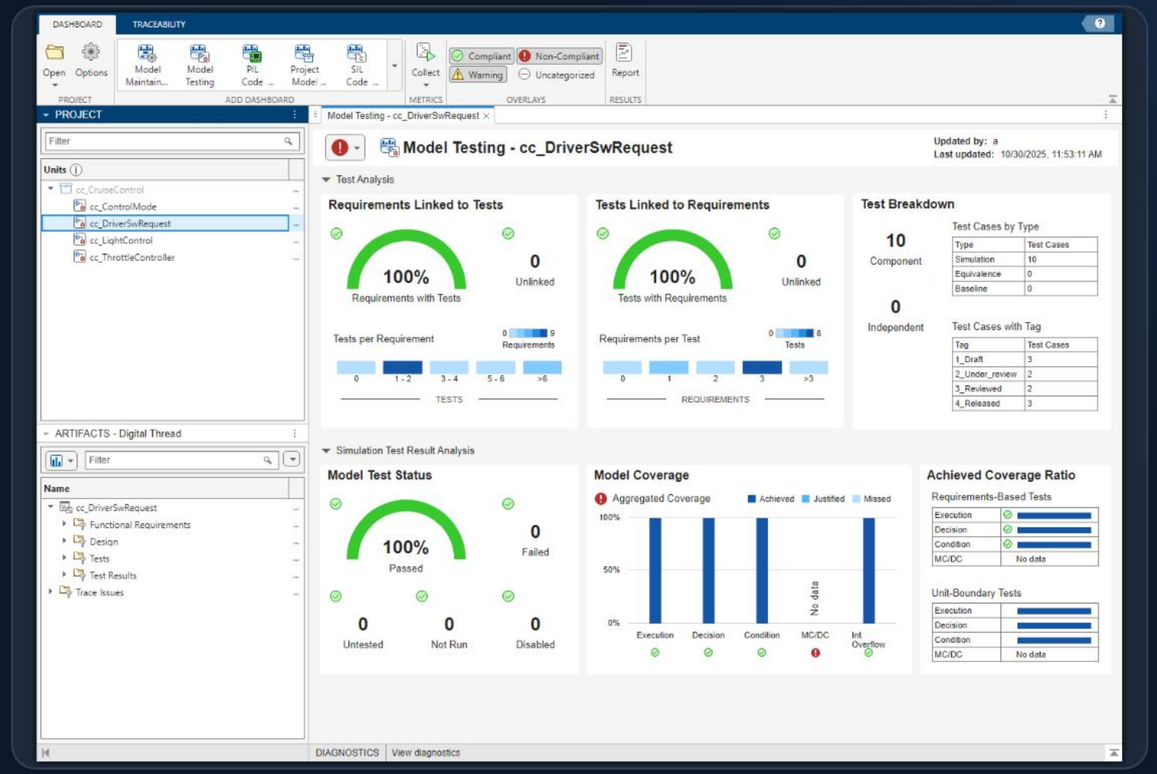
Promos & Validation

Promo Code Application: Valid codes apply discounts and record on booking; blank strings are treated as null without errors (`BK-04/05`).

Defensive Isolation: Referencing passengers owned by another account throws `PassengerNotFoundException` (`BK-07`).

Concurrency & Deadlock Prevention

- **Ascending ID Lock Ordering:** Seats are locked strictly in ascending ID order, preventing lock cycles across racing checkouts (`BK-01`).
- **Opposite Order Concurrency Test:** Tested in `BookingFlowIT.java:173` — simultaneous requests over overlapping seats neither deadlock nor double-book (`BK-09`).
- **Database Transaction Truth:** Real PostgreSQL Testcontainers verify pessimistic row locks under multi-threaded contention (`BK-10`).



Seat Hold & Expiration Mechanics

1. TTL Hold & Re-affirmation

Holds seat with expiration. Submitting a new hold by the same owner extends the TTL rather than throwing an error (`SH-03`).

2. Privacy-Safe Release

Releasing a hold owned by someone else is an idempotent **silent no-op**, preventing state discovery (`SH-06`).

3. Bulk Sweeper & Expired Reclaim

Expired holds are reclaimed on access or swept in bulk by `releaseExpiredHolds` without locking the table (`SH-09`).

Payment Idempotency & 3-D Secure

Payment Idempotency & Retries

Reference Replay: Replayed payment references return the original payment result without re-charging (`PAY-01/07`).

Decline & Retry: A declined card leaves the seat hold intact, allowing the customer to retry with a valid card (`PAY-06`).

3-D Secure Challenge Workflow

Pending Challenge: Cards requiring 3DS save payment as `PENDING_3DS` without settling the booking (`PAY-12`).

Challenge Settlement: Valid challenge code confirms payment & booking; invalid code declines payment (`PAY-13`).

Reschedule & Fare Settlement

Side-Effect-Free Quotes

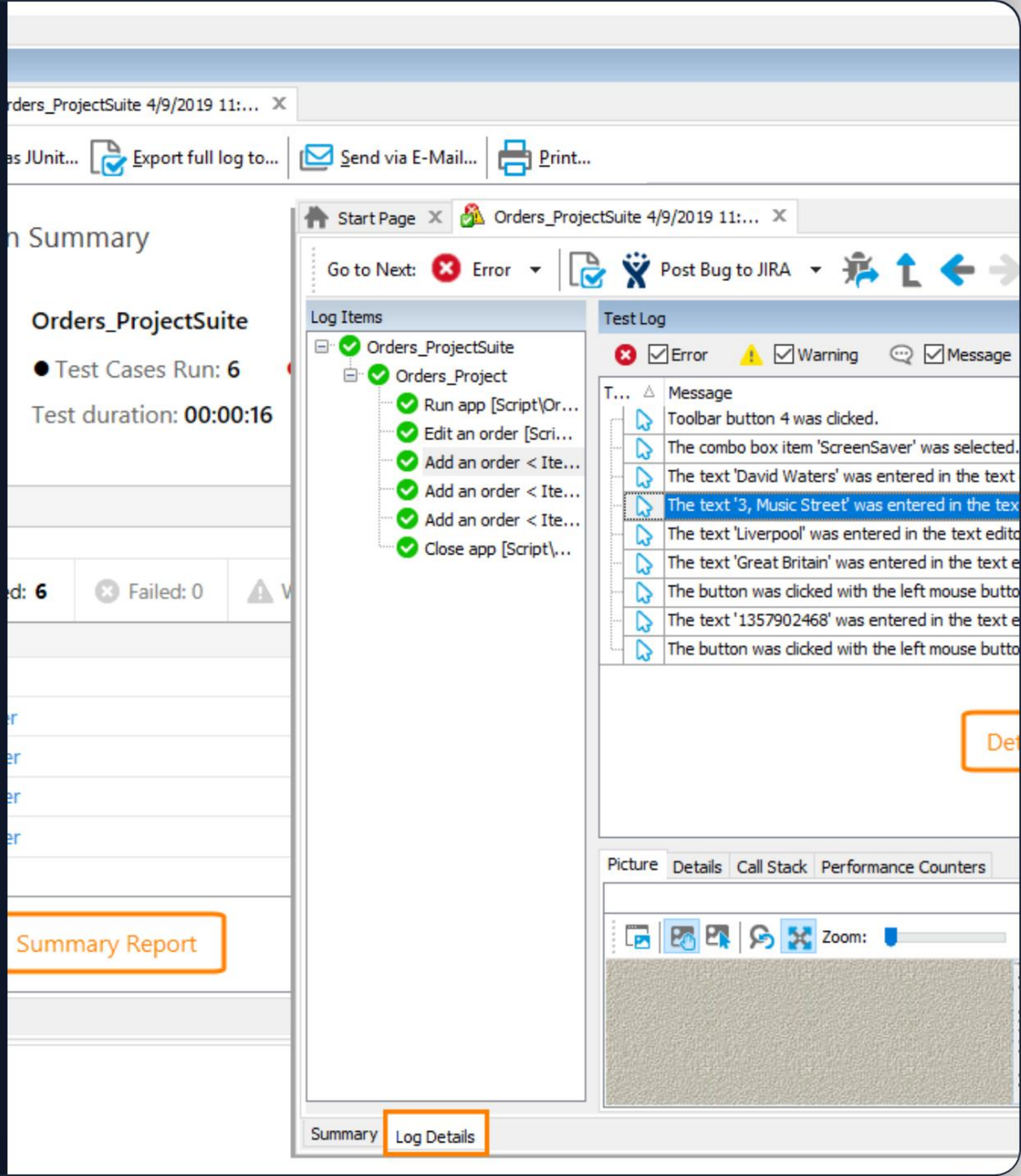
`previewReschedule` calculates fare differences and proximity rules without mutating booking state (`RS-01/02`).

Fare Top-ups & Credit Refunds

Upgrades collect top-up payments; downgrades reduce payment total and issue a `RESCHEDULE_CREDIT` refund (`RS-07/09`).

Proximity Guarding

Reschedule requests submitted too close to departure throw `CancellationNotAllowedException` (`RS-10`).



Refund Lifecycle & Agent Overrides

1. Customer Initiation

Creates a `PENDING` refund. Booking stays `CONFIRMED` and seats remain booked during support review (`RF-01`).

2. Support Approval

Marks refund `PROCESSED` , refunds payment, cancels booking, and releases seats for resale (`RF-07`).

3. Agent Fee Waiver

Support agents can apply fee waivers with a mandatory reason, persisting the waived delta (`RF-09`).

TICKETWAVE FUNCTIONAL TESTS VERIFIED

Questions & Discussion

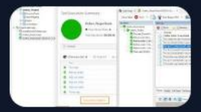
670 Tests Passing | 100% Financial Module Coverage Gate Enforced

```
cd backend && mvn clean verify
```

Image Sources

https://au.mathworks.com/help/slcheck/ref/model-testing-dashboard-complete.png

Source: www.mathworks.com



https://static1.smartbear.co/smartbearbrand/media/images/product/testcomplete/testreportinganalysis_mainbanner.png

Source: smartbear.com